# Aoji Tang

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## Education

**University of North Carolina at Chapel Hill (UNC)** *Chapel Hill, US*

MS in Statistics, Exchange Program *Aug 2024 - Dec 2025, Fall 2023*

- Thesis advisor: Dr. Michael J. O'Neill
- Thesis title: An inexact two-stepsize stochastic SQP (sSQP) method for nonlinear optimization problems with rank-deficient constraint Jacobians

**Fudan University** *Shanghai, CN*

BS in Mathematics & Applied Mathematics *Sep 2019 - Jun 2024*

- Thesis advisor: Dr. Qi Zhang
- Thesis title: Stationarity tests and change point detection on macroeconomic data and future market

## Research Experience

2025-now **Complexity Analysis on Constrained Optimization,** Research Assistant *UNC*

- Designed and analyzed an inexact two-stepsize sSQP algorithm for nonlinear constrained optimization under weak conditions.
- Derived first-known worst-case complexity bounds of sSQP method on constraint violation in the absence of constraint qualifications, whose order matches the existing optimum.
- Tested the algorithm on CUTEst and achieved lower infeasibility errors in less time than benchmark algorithms.

Fall 2024 **Universal Parameter-free Gradient-based Algorithm Analysis,** Individual *UNC*

- Extended the convergence result of Distance over Weighted Gradients (DoWG) algorithm to a strong stochastic gradient setting.
- Compared the performance of DoWG and three different types of gradient-based algorithms on deep learning tasks with Python.

Spring 2024 **Randomized Simplex Pivoting Rule,** Research Intern *UNC*

- Proposed and tested a new randomized pivoting rule of simplex method which avoids worst-case performance on Klee-Minty cubes with Matlab.
- Computed the expected pivoting times of the rule under a weighted discrete distribution.

2023-2024 **Dual-class Stock Price Analysis,** Research Intern *Fudan*

- Web-crawled repurchase data and financial reports from companies with dual class stocks, and ran regressions for interpretation.
- Proposed Repurchase Intensity as a feature to analyze and interpret the price difference between dual stock classes.

Spring 2023 **Undergraduate Thesis,** Author *Fudan*

- Reviewed and applied various time-series stationarity tests and change point detection tests on macroeconomic data and future markets and captured structural break in 2008 and 2020.

## Awards, Fellowships, & Grants

2025 **Master's Student Scholarship,** UNC STOR *$ 5,300*

**Student Travel Grant,** ICCOPT 2025 *$900*

Fall 2023 **Dean's List,** UNC (Exchange Program)

2023 **Professional Scholarship,** School of Mathematical Sciences, Fudan University *¥ 1,000*

2022 **Professional Scholarship,** School of Mathematical Sciences, Fudan University *¥ 1,000*

## Teaching Experience

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| Fall 2025 | **STOR 535: Probability for Data Science,** Instructional Assistant | *UNC* |
| Spring 2025 | **STOR 155: Introduction to Data Models and Inference,** Teaching Assistant | *UNC* |
| Fall 2024 | **STOR 415: Introduction to Optimization,** Teaching Assistant | *UNC* |
| Spring 2023 | **Probability Theory and Mathematical Statistics,** Teaching Assistant | *Fudan* |

## Project & Workshop Experience

Fall 2024 **Reproducible Data Analysis on Enron Corpus Data,** Group Leader *UNC*

- Cleaned email data and clustered employees into nine business departments with R.
- Conducted semantic analysis with ChatGPT API on the email content.
- Maintained a (Nature Methods) silver-standard reproducible github repository. [link]

Fall 2024 **Tree Canopy Detection with Deep Learning,** Group Leader *UNC*

- Applied objection detection and image segmentation models such as YOLOv11, U-Net and SAM to detect and segment tree canopies for better managing forest resources.
- Labeled and annotated raw remote sensing images of forest to enlarge dataset. Enhanced shadow features to improve detection accuracy by 10%.

Aug 2023 **Polymath Jr. 2023,** Member *Online*

- Studied data assimilation, optimal transportation and their connection with generative machine learning with group members in weekly discussion.

Fall 2021 **Consumer Behavior Experiment and Analysis,** Group Leader *Fudan*

- Designed an offline social psychological experiment about consuming behavior, discounts and advertisements, and conducted it on 30 subjects.

## Leadership & Extracurricular

2019-2021 **Fudan Art Troupe,** Director of Administration Dept. *Fudan*

- Supported business development of the university's graduation festival. Sponsorship added up to more than ¥180,000.
- Led a group of 20+ members on daily administrative work of the whole Troupe, managed to recruit more than 70 skilled new members.